VIP: Interacting with Sound and Space

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Lab for Interaction & Immersion
School of Music, College of Design
Georgia Institute of Technology

l42i.music.gatech.edu

Students join across class years and stay with the team over multiple semesters, taking on progressively larger parts of the work on interactive and immersive audio systems.

- **AFFILIATION:** College of Design (School of Music), Georgia Tech
- **COURSE:** VIP 2601 to 6603 (SO to GR)
- **GENERAL MEETING TIME:** TBA

BIKES

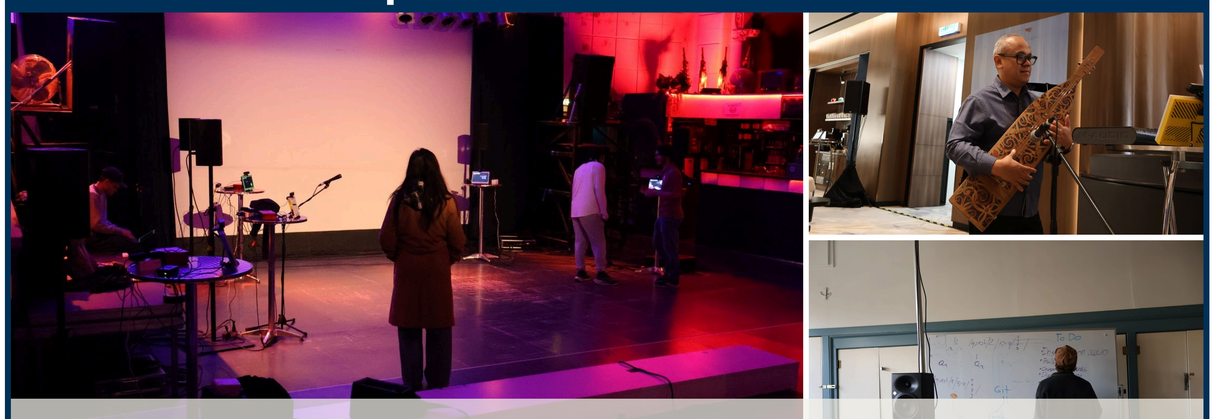


This project combines industrial design with all kinds of engineering. The four networked bikes with loudspeakers can be used for critical interventions in the urban space. Our latest version runs AI-supported camera tracking. Next step: new hardware design (the black edition).

MAJORS NEEDED

Industrial Design, Music Tech, Computer Science, Electrical Engineering, Public Policy, Literature, Media, and Communication, ...

MËSH and SuperMËSH



This project explores distributed systems for immersive multi-user interaction. MËSH has traveled a lot - across the US, to Europe and Asia - to create experiences for musicians and audiences. With SuperMËSH we are taking it to a next level with distributed multi-camera movement tracking.

MAJORS NEEDED

Computer Science, Computational Media, Electrical Engineering, Music Tech, ...

XR Music Interaction



Our lab explores how we can combine XR interfaces with 3D loudspeaker systems for immersive interactive experiences. This lets musicians and everyone move sound in space through augmented objects. We explore new methods for evaluating the interaction in creative contexts.

MAJORS NEEDED

Music Tech, Computer Science, Computational Media, Electrical Engineering, Industrial Design, ...

brain.wav



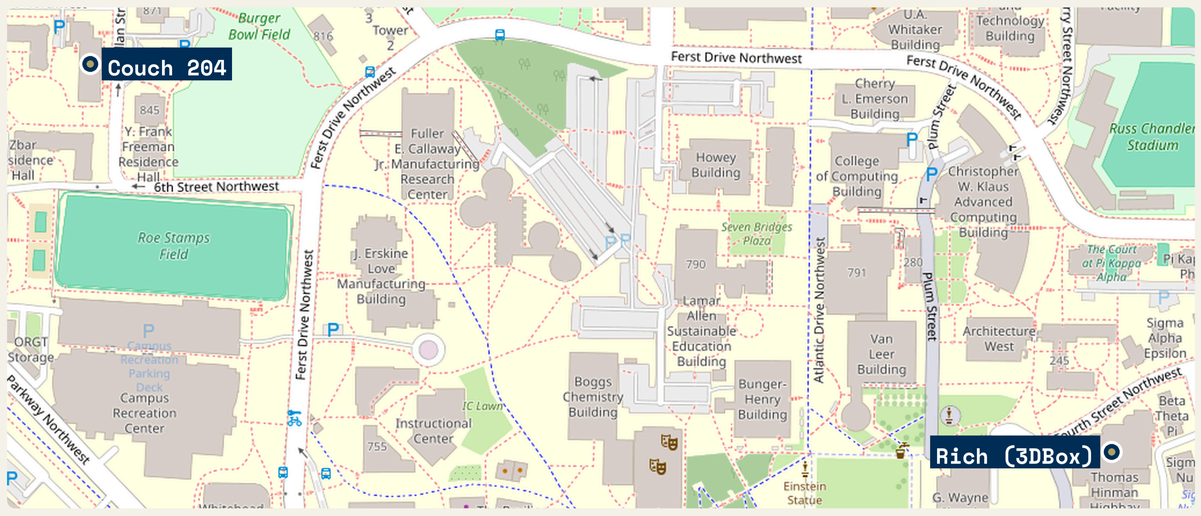
In collaboration with XXX we are exploring how we can use immersive loudspeaker systems to make fMRI data audible. This HCI research aims at tools for neuroscience researchers. With real-time input devices, they can navigate the recorded data, while hearing it.

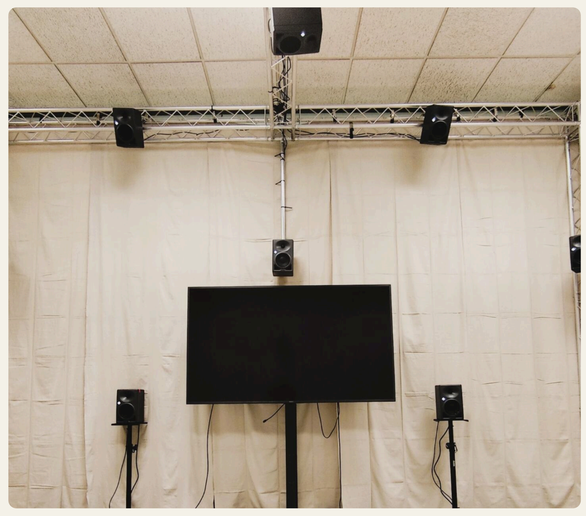
MAJORS NEEDED

Music Tech, Computer Science, Neuroscience, Biomedical Engineering, Psychology, ...

OUR SPACES







Couch 204
Main multi-purpose lab space with a 16-channel dome, desks and project areas.

Rich (3DBox)
Immersive sound system with 29 channels for research and artistic practice.